T503 Environment Biotechnology

Question No : 11 of 50 Marks: 1 (Budgeted Time

n the conventional activated sludge system, continuous arrival of food in the form of BOD to the aeration tank, leads to rapid and continuous bacterial growth. ll of the following are problems caused by increased bacterial growth, except:

Answer (Please select your correct option)

- The overall efficiency of the process is enhanced ✓
- Solid material leaves with the final effluent
- The solids do not settle well
- The secondary sedimentation tank becomes overloaded

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83:00
Time Left

11

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41

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T503 Environment Biotechnology MC

Question No : 21 of 50 Marks: 1 (Budgeted Time Min)

Unlike other marines species. _____are characteristically untroubled by biofouling and studies of their skin have established that its unique texture coupled with its inherent antimicrobial properties help keep them clean of fouling organisms.

Answer (Please select your correct option)

- sharks ✓
- dolphins
- sea-horses
- oysters

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21

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44

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BT503 Environment Biotechnology

Question No : 3 of 50 Marks: 1 (Budgeted Tim

Which plant is used as a model to protect from various agents?

Answer (Please select your correct option)

- Flowery plants
- Cotton plants
- Pea plants
- Tobbacco Plants ✓

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3

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41

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Which statement about chlorinated aliphatic hydrocarbons is correct?

Answer (Please select your correct option)

- They usually act as proton donors. ✓
- The ability to dehalogenate CAHs is found in very few microbes.
- Oxidative dehalogenation of CAHs is carried out by facultative aerobes.
- They may be converted into other products through a process known as dehalorespiration.

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Question No : 37 of 50 **Marks: 1 (S**

Which of the following approaches do not remove the pollutants but only immobilize the pollutants?

Answer (Please select your correct option)

- Rhizofilteration
- Bio-scouring
- Phytostabilization
- Biofilteration

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37

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38

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Question No : 10 of 50

The airflow in the submerged aerated biofilter is always _____ while the liquid flow can be _____.

Answer (Please select your correct option)

- upflow, downflow
- downflow, upflow
- downflow, downflow
- upflow, stagnant

BT503 Environment Biotechnology

Question No : 1 of 50 Marks: 1

Deliberate transfer of a gene from one organism to another is called.....

Answer (Please select your correct option)

- Foreign gene ✓
- Reported gene
- Supresser gene
- Oncogene

Start Time: 6:11 PM

Question No : 2 of 50 Marks: 1

Western Blotting technique is used for the detection of.....

Answer (Please select your correct option)

- DNA
- RNA
- Protein ✓
- none of these

Which statement about chlorinated aliphatic hydrocarbons is correct?

Answer (Please select your correct option)

- They usually act as proton donors.
- The ability to dehalogenate CAHs is found in very few microbes.
- Oxidative dehalogenation of CAHs is carried out by facultative aerobes.
- They may be converted into other products through a process known as dehalorespirat

BT503 Environment Biotechnology

Question No : 24 of 50 Marks: 1 (Budgeted T

Among the first synthetic organic chemicals to create environmental problems were _____.

Answer (Please select your correct option)

- Detergents
- Plastics
- Pesticides
- CFCs

Start Time: 5:01 PM

82:00

24

plants containing the novel DNA grow, a gene coding is frequently for------

Answer (Please select your correct option)

- weed-killer
- antibiotic
- Resistance
- all of above ✓

Question No : 4 of 50

For land based applications, the soil types best suited to biotechnological interventions are _____

Answer (Please select your correct option)

- Sand and Gravel ✓
- Sand and Organic Matter
- Sand and vermiculite

BT503 Environment Biotechnology

Question No : 13 of 50

Which of the following is NOT a possible destination of the pollutants in soil?

Answer (Please select your correct option)

- appearance in surrounding air
- appearance in the underground water

The presence of metals in environment may be detected visually by the use of bacteria. One such example is *Vogesella indigofera* that gives an idea of metal contamination by producing _____

Answer (Please select your correct option)

- coloured pigments
- hydrolytic enzymes
- oxygen gas

T503 Environment Biotechnology

Question No : 19 of 50

Which statement about bioscrubber is FALSE?

Answer (Please select your correct option)

- They are appropriate for hydrophilic compounds such as methanol
- They are very efficient for removal of odours
- They use water for the removal of odourous compounds
- They are useful for the removal of toxic metals ✓

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T503 Environment Biotechnology

Question No : 39 of 50 Marks: 1 (Budgeted)

Which kind of techniques are suitable for phytoremediation ....

Answer (Please select your correct option)

- In situ ✓
- Ex situ
- Physical
- Bio-leaching

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39

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 4

BT503 Environment Biotechnology

Question No : 9 of 50 Marks: 1 (Budgeted Time M

The _____ products of a pollutant may be more harmful than the original pollutant itself. Therefore it is important to understand the chemistry of the pollutant

Answer (Please select your correct option)

- breakdown
- concentrated
- formulated

BT503 Environment Biotechnology

Question No : 38 of 50 Marks: 1 (Bud

How much arsenic can be accumulated by using Pteris vittata?

Answer (Please select your correct option)

- 20-30%
- 5-10%
- 2-3%
- 60-70%

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38

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38

Answer (Please select your correct option)

- adsorption
- digestion
- hydrolysis

BT503 Environment Biotechnology

Question No : 27 of 50 **Marks: 1**

Dioxins have many congeners. The most toxic is _____, which is thought to be the most poisonous synthetic chemical.

Answer (Please select your correct option)

- DDT
- TCDD
- CFC
- PPE

Start Time 5:01 PM

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27

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 3

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BT503 Environment Biotechnology

Question No : 40 of 50 — Marks: 1 (Budgeted Time

Which of the following methods are best suited to instances of relatively localized pollution within a site, typically in 'hot-spots' of medium to relatively high concentration which are fairly near to the surface.

Answer (Please select your correct option)

- Solidification
- Vitrification
- In situ
- Ex situ

Start Time 5:01 PM

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40

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41

for by Amaan Khan

BT503 Environment Biotechnology

Question No : 4 of 50

Which of the following factors has no effect on the execution and implementation of environmental biotechnology?

Answer (Please select your correct option)

- Location
- Economics
- Cultural norms
- Health of the general population

AMAAN KHAN

BT503 FINAL Terms Current M.C.Q Spring 2020

1- Which of the following methods are suited to instance of relatively localized pollution? **Ex.situ**

2- Dioxins have many congeners. The most toxic is --------- which is thought to be most poisonous synthetic chemicals. **TCDD**

3- The presence of metals in environment may be detected visually by the use of bacteria. One such example is Vogesella Udigofera that gives an idea of metal contamination. **Colored pigments**

4- Which of the following is NOT a possible destination of the pollutants in soil. **Appearance in surrounding air**

5- For land based application the soil type best suited to biotechnological interventions are ---. Sand and gravel

6- Plants containing the noval DNA grow a gene coding is frequently for-------. **All of above**

7- Deliberate transfer of a gene from one organism to another is called --------. **Foreign gene**

8- Western blotting technique is used for the detection of --------. **Protein**

9- Immobilization refers to the removal of contaminants typically metals by means of ------- by various microbes or plant species**. Adsorption**

10- How much arsenic can be accumulated by using Pteris vittata. **2-3%**

11- The ----- products of a pollutant may be more harmfull than the original pollutant itself.

12- Therefore it is important to understand the chemistry of the pollutants. **Breakdown**

13- Among the first synthetic organic chemicals to create environmental problems were ----. **Detergents**

14- Which of the following factors has no effect on the execution and implementation of environmental biotechnology? **Health of general population**

15- The airflow in the submerged aerated biofilter is always----- while the liquid flow can be ---- **Upflow and downflow**

16- Which statement about chlorinated aliphatic hydrocarbon is correct? **They usually acts as proton donors**

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17- Which plant is used as a model to protect from various agents? **Tobacco plant**

18- In the conventional activated sludge system, continuous arrival of food increased bacterial growth except. **The overall efficiency of the process is emhanced**

19- Unlike other marines species ------ are charectically untroubled by biofouling. **Sharks**

20- Which kind of technique are suitable for phytoremediation? **In-situ**

21- Which statement about bioscrubber is False? **They are useful for the removal of toxic metals.**

22- The term endophytes is usually used to denote ----- microorganisms. **Non pathogenic**

22- Which of the following have innate ability to degrade the hydrocarbon contaminants through metabolic action and biosurfactant production? **Hudrocarbonolastic bacteria**

23- What is not true about bio scrubbers? **Remove toxic metal**

24- Facultative ponds which receive raw sewage is also known as -------. **Primary ponds**

25- The bio scrubber is not itself truly a biological treatment system but rather a_____. **Highly efficient method of removing odour components by dissolving them**

26- Biodiesel is not derived from --------. **Algae**

27- An aquatic plants_____ **Macrophyte**

28- Important factors to be considered when evaluating and selecting unit operations and processes except _____ **Sterilization**

29- Dioxins released as by products in a number of chemical process. **Do not pose threats to environments**

30- Non specific enzymes are --------. **Oxygenases**

31- Waste water is applied to the soil supplying water and nuterients necessary for plant growth. Which is not true? **Generation of aerosols**

32- Which is not used regarding desulphurization of coal? **Psychrophile**

33- Suitable temperature is _____ **20-30 degree Celsius**

34- What is false about collection stage? **This is stage when solid waste is transported to the final disposal site**

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35- Which of ------ does not use to make vessal of biofilters? **Formic sheet**

36- Preliminary treatment involves the removal of _____ **Coarse solid**

37- In bacteria genetic transfer of materials occurs through_____ **Plasmids**

38- Wolffia globose is an important tool fo_____ **cadmium**

39- Optimum PH for local environment conditions lies in the range_____ **6.5 -7.5**

40- ------ uses high temperature for fuse contaminated material. **Vitrification**

41- A step in the development of biosensors is? **The immobilization of biological component at the transducer surface.**

42- Association of fungi with roots of vascular plants. **Mycorhizzae**

43- The ability of a single compound to be degraded ca be affected by the presence of other contaminants. **Cometaboliusm**

44- Thermophiles thrive at temperature above? **45 degree Celsius**

45- -------- are compounds which are manmade. **Xenobiotics**

46- Natural estrogen is deactivated in human by -------. **Glucuronidation**

47- Collection of total genomic DNA from organisms_____ **Genomic libraries**

48- ------- refers to how pollutants enter a food chain. **Bioaccumulation**

49- UASB reactor stands for --------- reactors. **Upflow anaerobic sludge blanket**

50- False statement about on site disposal of waste------. **Waste volumes are larger**

51- Explosives characterized by the presence of ------- group. **Nitro**

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BT503_Current Final Term Subjective Papers Spring 2020

1. Define Halophiles. 2
2. Define psychrophiles. 2
3. Define Biosensors. 2
4. Define co-metabolism. 2
5. How microbes are poly cyclic aromatic compounds. 2
6. Write two advantages of the conventional biological sludge dewatering process. 2
7. How plants are modified as bioindicator. 2
8. Mention two Natural sludge dewatering processes. 2
9. How quality of fruit should improve 2
10. Write two sources of dioxins? 2
11. What are xenobiotic? 2
12. Ex-situ technique? 2
13. What is Glyphosate 2
14. Define Bioscrubber 2
15. What is Sustainable bioremediation 2
16. Why DDT was banned? 2
17. Write two methods of Detergents bioremedies. 2
18. Consequences of OECD 2001 about global Biotechnology? 2
19. Define phytodegradation and name its forms? 2
20. What are living conditions of extremophiles? give examples 2
21. Difference between bioenhancement and biogumentation. 2
22. Difference between natural and conventional sludge dewatering process. 3
23. Difference between physical and mechanical sludge dewatering process. 3
24. Name enzyme, solution and equipment's used in the genetic engineering. 3
25. There classes of biosensor and two advantages and disadvantages of immunsensor. 3
26. Enlist the types of solid waste. 3
27. Phytoplankton? 3
28. How quality of crops should enhance? 3
29. Write 3 oxidation products of TNT. 3
30. Process of thickening of sludge. 3
31. Write features of phenols that make it pollutant? 3
32. What is T-DNA? 3
33. What is Environmental biotechnology 3
34. methods to get a good quality crop 3
35. biological remediation. 3
36. Advantages of association of fungi with plants 3
37. With help of named example discuss how whole organisms can be used the measure potential biological Toxicity
of water soil sample? 3

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38. Describe importance of anaerobic zone in biological process of phosphorus removal in activated sludge treatment?3
39. Describe natural process to remove pathogenic environment. 5
40. Environmental concern to the release of the hormone into environment. 5
41. Role of filter medium in biofilter also name the compound of the filter medium. 5
42. Five impact of chlorobenzene & dioxin on human health. 5
43. write a note on cDNA libraries 5
44. Write the five properties of soil. 5
45. Write five advantages of Biofilter 5
46. Explain Vitrification methods 5
47. Why DDt banned reason of its banned although it is useful method for removal of insects?5
48. Genetic modification a popular plant is used for removal of mercury. describe how? 5
49. Factors affecting the techniques of Insitu and exsitu. 5
50. What are applications of biosensors in detection of biocides? 5
51. Explain role of enzyme, solution & equipment in genetic engineering? 5
52. Explain genomic libraries? 5
53. solid waste steps. 5
54. What are environment factor that effect bioremediation briefly describes? 5
55. List any five source of solid waste? 5
56. What are PCBS? Which biosensors detect them? 5
57. Mention some metallic coagulant for sludge dewatering? 5
58. Phytovolatilisation 10
59. Dental not sludge treatment stage before final disposal steps. 10
60. Describe biosensors single transduction and bio recognition principle. 10
61. Write the procedure of Sludge treatment 10
62. Explain western and Northern blotting. 10
63. Cloning Vector. 10
64. Briefly describe different categories of Solid waste 10
65. What are risks of Solid Wastes regarding transfer of disease? 10

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